



Ext 2

NORTH SYDNEY BOYS HIGH SCHOOL

2010
ASSESSMENT TASK 4

Mathematics

Extension 2

General Instructions

- Working time – 50 minutes
- Write in the booklet provided
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question
- Each new question is to be started on a new page.

- Attempt all questions

Class Teacher:

(Please tick or highlight)

- ☐ Mr Barrett
- ☐ Mr Trenwith
- ☐ Mr Weiss

Student Number:

(To be used by the exam markers only.)

Question No	1	2	3	4	5	Total	Total
Mark	$\frac{5}{5}$	$\frac{6}{6}$	$\frac{7}{7}$	$\frac{9}{9}$	$\frac{11}{11}$	$\frac{38}{38}$	$\frac{100}{100}$

Question 1 (5 marks)

Marks

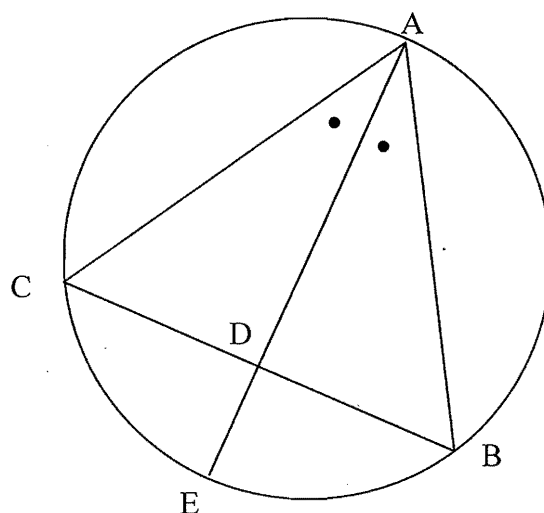
A sequence U_n is defined as $U_{n+2} = 6U_{n+1} - 5U_n$.

5

$U_1 = 2$ and $U_2 = 6$

Use mathematical induction to prove that $U_n = 5^{n-1} + 1$

Question 2 (6 marks)



In the diagram above AD, extended to E is the bisector of $\angle BAC$.

Copy this diagram into your answer book.

a) Prove that $\triangle ABE$ and $\triangle ADC$ are similar

2

b) Show that $AB \times AC = AD \times AE$

1

c) Prove that $AD^2 = AB \times AC - BD \times DC$

3

Question 3 (7 marks)

a) Find the term independent of x in $\left(x^2 + \frac{1}{2x}\right)^{18}$ 3

b) Show that $\sum_{r=0}^n \frac{{}^nC_r}{r+1} = \frac{2^{n+1} - 1}{n+1}$ 4

Question 4 (9 marks)

A mass of m kilograms falls from a stationary balloon at a height above the ground. It experiences air resistance during its fall equal to mkv^2 , where v is its speed in metres per second and k is a positive constant.

Let x be the distance in metres of the mass from the balloon measured positively as it falls.

a) Show that the equation of motion of the mass is $\ddot{x} = g - kv^2$, 1
where g is the acceleration due to gravity.

b) Show that $v^2 = \frac{g}{k}(1 - e^{-2kx})$ 4

c) Find the terminal velocity of the mass. 2

d) Find the distance fallen when the mass reaches half of its terminal velocity. 2

TURN OVER FOR QUESTION 5

Question 5 (11 marks)

Suppose $a > 0, b > 0$ and $c > 0$

a) Prove $a^2 + b^2 \geq 2ab$ 2

b) Hence prove $a^2 + b^2 + c^2 \geq ab + bc + ca$ 2

c) Given that

$$a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$$

prove that $a^3 + b^3 + c^3 \geq 3abc$ 2

d) If x, y, z are > 0 ,

Make a suitable substitution into part iii) to show that

$$x + y + z \geq 3(xyz)^{\frac{1}{3}}$$
 1

e) Suppose $(1 + x)(1 + y)(1 + z) = 8$, prove that $xyz \leq 1$ 3

END OF EXAMINATION

Q1

5 marks

Step 1

$$n=1 \Rightarrow u_1 = 5^{1-1} + 1 = 2 \quad \text{True} \quad |$$

$$n=2 \Rightarrow u_2 = 5^{2-1} + 1 = 6 \quad \text{True} \quad |$$

Step 2Assume true for $1, 2, 3, \dots, n$

$$u_n = 5^{n-1} + 1 \quad |$$

$$\text{Consider } u_{n+1} = 6u_n - 5u_{n-1}$$

$$= 6(5^{n-1} + 1) - 5(5^{n-2} + 1)$$

$$= 6 \times 5^{n-1} + 6 - 5^{n-1} - 5 \quad |$$

$$= 5 \times 5^{n-1} + 1$$

$$= 5^n + 1$$

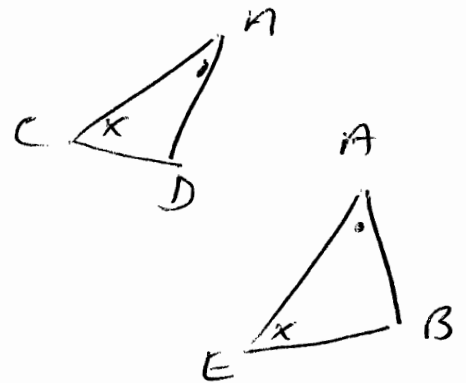
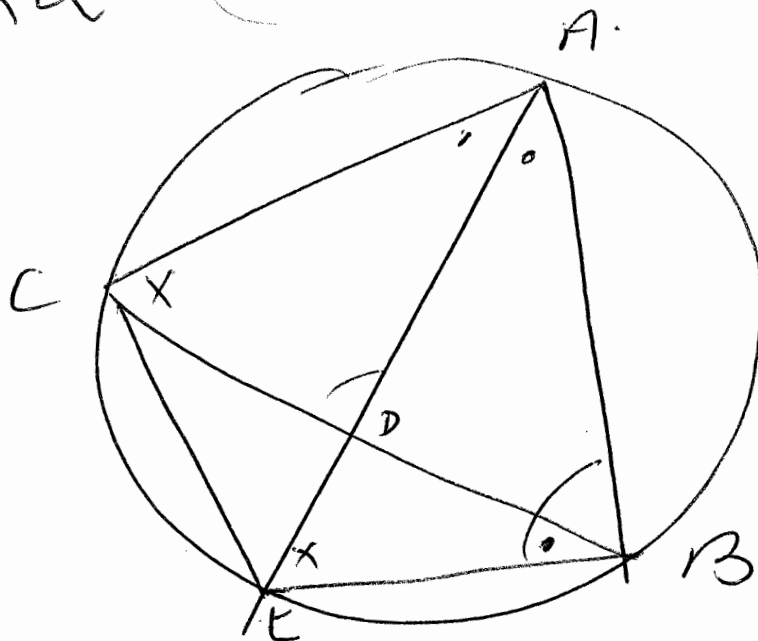
$\therefore$ If true for u_n it is true for u_{n+1}

As it is true for $n=1, 2, 3, \dots$

it is true for all n by

Mathematical Induction.

Q2. (6 marks)



i) In Δ s ΔABE and ΔAPC

$$\angle EAB = \angle DAC \quad (\text{given})$$

$$\angle AEB = \angle ACD \quad (\text{angle standing on the same arc AB})$$

$$\therefore \Delta ABE \sim \Delta ADC \quad (\text{AAA})$$

$$\text{ii) } \frac{AB}{AE} = \frac{AD}{AC} \quad (\text{Sides in proportion})$$

$$\therefore AB \times AC = AD \times AE \quad (1)$$

iii) Prove $AD^2 = AB \times AC - BD \times DC$ (2)

$$\text{Now } AD \times DE = BD \times DC \quad (3) \quad (\text{Intersecting chords})$$

$$\text{Consider } AB \times AC - BD \times DC$$

Subs (1) + (3)

$$AD \times AE - AD \times DE = DC$$

$$AD(AD + DE) - AD \times DE = DC$$

$$AD^2 + AD \times DE - AD \times DE = AD^2$$

$$3) T_{13} = \frac{18!}{(12!)^2} \cdot \frac{1}{2^{12}}$$

$$\Rightarrow \frac{20^{36-3k}}{2^k} = 1$$

$$\text{w. } 2^{36-3k} = 1$$

$$36-3k=0$$

$$k=12$$

$$\therefore T_{13} = \frac{18!}{(12!)^2} \times \frac{1}{2^{12}}$$

$$= \frac{18!}{(12!)^2} \times \frac{1}{2^{12}}$$

$$= \frac{4641}{1024}$$

$$1024$$

$$3) \quad (1+x)^n = \sum_{r=0}^n \binom{n}{r} x^r$$

$$\text{LHS} = \frac{\binom{n}{0}}{1} + \frac{\binom{n}{1}}{2} + \frac{\binom{n}{2}}{3} + \dots + \frac{\binom{n}{r}}{r+1} + \dots + \frac{\binom{n}{n}}{n+1}$$

$$\text{Consider } \int (1+x)^n dx$$

$$= \int 1 + \binom{n}{1}x + \binom{n}{2}x^2 + \dots + \binom{n}{r}x^r + \dots dx$$

$$= \frac{(1+x)^{n+1}}{n+1} + C = \sum_{r=0}^n \frac{\binom{n}{r}}{r+1} x^{r+1} + \dots$$

$$x=0 \Rightarrow$$

$$\frac{1}{n+1} + C = 0 \quad \text{as} \quad C = -\frac{1}{n+1}$$

$$\frac{(1+x)^{n+1}}{n+1} - \frac{1}{n+1} = \sum_{r=0}^n \frac{\binom{n}{r}}{r+1} x^{r+1}$$

$$\text{let } x = 1$$

$$\frac{2^{n+1}}{n+1} - \frac{1}{n+1} = \sum_{r=0}^n \frac{\binom{n}{r}}{r+1}$$

Q4.

(mark)

↓
mg

↑
mkv²

a) $m\ddot{x} = mg - mkv^2$

$\therefore \ddot{x} = g - kv^2$

b) $\therefore \ddot{x} = v \frac{dv}{dx} = g - kv^2$

$$\frac{dv}{dx} = \frac{g - kv^2}{v}$$

$$\frac{dx}{dv} = \frac{v}{g - kv^2}$$

$$\int dx = \int \frac{v dv}{g - kv^2}$$

$$x + c = -\frac{1}{2k} \log(g - kv^2)$$

$$x=0, v=0 \Rightarrow c = -\frac{1}{2k} \log g$$

$$x - \frac{1}{2k} \log g = -\frac{1}{2k} \log(g - kv^2)$$

$$x = \frac{1}{2k} \log \left(\frac{g}{g - kv^2} \right)$$

$$\log \left(\frac{g}{g - kv^2} \right) = 2kx$$

$$e^{2kx} = \frac{g}{g - kv^2}$$

$$\frac{g - kv^2}{g} = e^{-2kx}$$

$$g - kv^2 = g e^{-2kx}$$

$$kv^2 = g - g e^{-2kx}$$

$$v^2 = \frac{g}{k} (1 - e^{-2kx})$$

$$A \quad c) \quad \ddot{x} = 0 = g - \kappa v^2$$

$$v^2 = \frac{g}{\kappa}$$

$$\therefore v = \sqrt{\frac{g}{\kappa}}$$

$$d) \quad v = \frac{1}{2} \sqrt{\frac{g}{\kappa}}$$

$$v^2 = \cancel{\frac{g}{\kappa}} \frac{g}{4\kappa}$$

$$\frac{g}{4\cancel{\kappa}} = \frac{g}{\cancel{\kappa}} (1 - e^{-2\kappa x})$$

$$\frac{1}{4} = 1 - e^{-2\kappa x}$$

$$e^{-2\kappa x} = \frac{3}{4}$$

$$-2\kappa x = \ln\left(\frac{3}{4}\right)$$

$$x = \frac{1}{2\kappa} \ln\left(\frac{4}{3}\right)$$

$$(a-b)^2 = a^2 - 2ab + b^2 \geq 0$$

a) $a^2 + b^2 \geq 2ab.$

b) $a^2 + b^2 \geq 2ab$ ①

$$b^2 + c^2 \geq 2bc$$
 ②

$$c^2 + a^2 \geq 2ac$$
 ③

$$\textcircled{1} + \textcircled{2} + \textcircled{3}$$

$$2a^2 + 2b^2 + 2c^2 \geq 2ab + 2bc + 2ac$$

$$\therefore a^2 + b^2 + c^2 \geq ab + bc + ca$$

c) Since $a, b, c > 0$

it follows $(a+b+c) > 0$

and from b) $a^2 + b^2 + c^2 > ab + bc + ca.$

$$\therefore (a+b+c)(a^2 + b^2 + c^2 - ab - bc - ca) > 0$$

$$\therefore a^3 + b^3 + c^3 - 3abc \geq 0$$

$$a^3 + b^3 + c^3 \geq 3abc$$

d) $a^3 = x \quad b^3 = y \quad c^3 = z$

$$a = x^{1/3} \quad b = y^{1/3} \quad c = z^{1/3}$$

$$\therefore x + y + z = 3(xy z)^{1/3}$$

$$d) \quad 8 = (1+x)(1+y)(1+z)$$

$$= (1+y+xy)(1+z)$$

$$= 1+z+y+y^2+x+xz+xy+xy^2$$

$$= 1+x+y+xy+xz+y^2+xy^2$$

$$\text{Now } x+y+z \geq 3(xy^2)^{1/3}$$

$$xy+xz+y^2 \geq 3(x^2y^2z^2)^{1/3} = 3x^{2/3}y^{2/3}z^{2/3}$$

$$\geq$$

$$8 \geq 1 + 3(xy^2)^{1/3} + 3(xy^2)^{2/3} + xyz$$

~~$$\text{let } u = (xy^2)^3$$~~

~~$$8 \geq 1 + 3u^{1/3} + 3u^{2/3} + u^3$$~~

~~$$\geq (1+u)^3$$~~

~~$$2 \geq 1+u$$~~

~~$$u \leq 1$$~~

~~$$xy^2 \leq 1$$~~

$$\text{let } u = xyz^{1/3}$$

$$8 \geq 1 + 3u + 3u^2 + u^3$$

$$8 \geq (1+u)^3$$

$$2 \geq 1+u$$

$$u \leq 1$$

$$xyz^{1/3} \leq 1$$

$$xyz \leq 1$$